

SUMMARY

Results-driven Software Engineer with deep expertise in Machine Learning, specializing in building fault-tolerant distributed systems and state-of-the-art models. Skilled at optimizing system performance, analyzing large-scale infrastructure issues, and delivering automated solutions that enhance operational efficiency. Proven ability to master and apply cutting-edge technologies while leveraging analytical expertise, coding proficiency, and systems thinking to solve complex reliability challenges.

EDUCATION

- **THE UNIVERSITY OF ADELAIDE**
MASTER OF AI AND MACHINE LEARNING
Sep 2022 - Dec 2024
Adelaide, Australia
- **Kazakh-British Technical University**
BACHELOR OF ENGINEERING IN COMPUTER SOFTWARE ENGINEERING
Sep 2017 - May 2021
Almaty, Kazakhstan

SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Transformers, Hugging Face, XGBoost, LightGBM, BERT, GPT, CNNs, RNNs, Transfer Learning
- **Data Science & Analysis:** NumPy, Pandas, Feature Engineering, Data Analysis, Research Methodology, Statistical Analysis, Data Visualization (Matplotlib, Seaborn)
- **Frameworks & Web Technologies:** React.js, Node.js, Django, Spring Boot, REST APIs, GraphQL
- **Databases & Storage:** PostgreSQL, MongoDB, SQL, Amazon S3
- **Cloud & DevOps:** AWS, Azure, Jenkins, Kafka, Windows Server

EXPERIENCE

- **Altyn Bank JSC (51% owned by China CITIC Bank Corporation)**
AI / LLM Engineer — Agentic Systems & Automation
May 2025 – Present
Almaty, Kazakhstan
 - Designed and implemented an **agentic document intelligence system** for the bank, automating ingestion, OCR, classification, and structured data extraction from financial documents (contracts, payment orders, invoices).
 - Built end-to-end pipelines using **Python, FastAPI, Docker, and Redis**, integrating OCR engines (PaddleOCR, LLM-based OCR) with **LLM-driven agents** for reasoning, validation, and error recovery.
 - Developed autonomous agents that transform unstructured documents into **Oracle-compatible XML payloads**, enabling direct integration with core banking **PL/SQL systems** and reducing manual processing time by **60%+**.
 - Implemented **queue-based and event-driven architectures** for high-volume document processing, ensuring reliability, auditability, and compliance with internal banking controls.
 - Integrated email and enterprise communication channels (IMAP/SMTP, Microsoft Graph) for automated document intake, routing, and status tracking.
 - Collaborated closely with compliance, operations, and IT teams to align AI-driven automation with **regulatory, security, and audit requirements** in a large financial institution.
- **SPOTFAKE.AI** 
Chief Product Officer & Full-Stack Developer
June 2024 - May 2025
Adelaide, Australia
 - Built and optimized cloud infrastructure using **AWS services** (EC2, S3, Lambda, RDS), achieving **40% faster load times** through code splitting, lazy loading, and CDN optimization strategies.
 - Developed internal platform tools including a high-performance CRM system using Python FastAPI backend and **Next.js/React.js frontend**, serving active users with **99.9% uptime**.
 - Designed developer-facing APIs and integrated FastAPI with **5+ client systems**, implementing comprehensive testing frameworks (stress testing, integration testing, gateway testing) to ensure reliability and security at scale.
 - Architected and optimized **Python-based data pipelines** for ML infrastructure, improving model accuracy by **25%** and reducing processing time by **35%** through efficient data workflow design.
 - Applied product thinking to technical decisions, balancing business needs with platform scalability and developer experience considerations.
- **THE UNIVERSITY OF ADELAIDE** 
RESEARCH & TEACHING ASSISTANT - AI/ML
March 2023 - Dec 2024
Adelaide, Australia
 - Demonstrated strong **developer advocacy and consulting skills** by mentoring **200+ students** across **6 semesters**, translating complex technical concepts into accessible learning experiences with **4.7/5 satisfaction rating**.
 - Developed **Python-based teaching tools and automation scripts** for grading workflows, reducing manual processing time by **50%** for a team of **8+ tutors**.
 - Applied customer empathy and consulting abilities to understand diverse learning needs, creating tailored educational pathways that improved student performance and engagement.
- **GoCrowd.io** 
Frontend Lead Developer & Platform Engineer
Jan 2022 - Dec 2022
Astana, Kazakhstan
 - **Led platform engineering initiatives** for a team of **5 developers**, building internal developer tools and frameworks that accelerated feature delivery by **40%** and established engineering best practices.
 - Architected and maintained **Kubernetes-based infrastructure** on **AWS cloud environment**, deploying containerized services with **sub-200ms response times**.

- Consulted with **3 product designers** and cross-functional teams to deeply understand user workflows, translating **50+ requirements** into scalable, developer-friendly platform capabilities.
 - Implemented comprehensive **observability and monitoring infrastructure** using industry-standard tools, establishing automated health checks and performance benchmarking that reduced critical incidents by **60%** and improved system reliability to **99.5% for 1000+ daily users**.
 - Applied **infrastructure-as-code principles** and configuration management to ensure platform consistency, implementing fault-tolerant design patterns including circuit breakers and retry mechanisms that reduced downtime by **70%**.
- **INDUSTRIAL AND COMMERCIAL BANK OF CHINA**  Feb 2021 - Dec 2021
Almaty, Kazakhstan
SOFTWARE ENGINEER
 - Developed enterprise applications using **Python, Java, and Spring Boot**, delivering **15+ platform features** focused on internal tooling and workflow automation.
 - Built an **internal management platform** serving **50+ employees**, applying platform engineering principles to streamline operational workflows and reduce manual processing time by **30%**.
 - Engineered **Python automation tools** that revolutionized daily reporting workflows, achieving **340% improvement** in accuracy and eliminating **20 hours/week** of manual effort through intelligent process automation.
 - Collaborated with internal stakeholders to understand pain points and requirements, translating business needs into scalable technical solutions.

ACADEMIC RESEARCH PROJECTS

- **WebCryptoAgent** Nov 2025 - Jan 2025
AIGEEKS GROUP
 - Conducted experimental research on memory vs stateless LLM trading agents, demonstrating performance gains from experience replay-based memory systems.
- **MAMBA STOCK** May 2024 - Oct 2024
Australian Institute of Machine Learning
 - This study introduces MAMBAStock, a novel approach based on the recently proposed Selective State Space (S6) model, MAMBA. MAMBAStock is designed to swiftly capture stock features and predict prices with competitive accuracy, under Dr. Jinan Zou from Australian Institute of Machine Learning.
- **Debiasing Hospital Review** Jan 2023 - Dec 2023
Australian Institute of Machine Learning
 - Initiated and currently leading a groundbreaking research project in collaboration with Dr. Weitong Chen, targeting the pervasive issue of gender bias within NLP algorithms. This ambitious endeavor, commenced in March 2023, is dedicated to identifying, analysing, and mitigating gender bias in NLP systems.

REFERENCES

Alikhan Okas (Fellow student)
SWE-SRE for Google Search
Google (Mountain View, USA)
alikhhan.okas@gmail.com

Angus Williamson (Colleague)
CEO
SPOTFAKE.AI
angus.williamson@spotfake.ai

Adilkhan Kozhakhmetov (Teammate in previous project)
Software Engineer
Google (London, UK)
adilkhan.kozhakhmetov@gmail.com

Dr. Tim Chen (Supervised me during my role as a Teaching Assistant)
Senior Lecturer
The University of Adelaide
tim.chen@adelaide.edu.au